

GDC Sandbox Workshop

Lab Guide

The goal of the workshop is to impart participants with an understanding of GDC sandbox and practical knowledge to deploy workloads. To ensure a smooth experience during the workshop, attendees are requested to install the necessary developer tools beforehand.

The lab section of the workshop is specifically designed for developers and requires these tools to complete the assigned tasks.

Pre- Requisites:

Bring Your Own Device with Admin access or have access to a **Virtual Desktop** with Admin access.

Developer Tools:

1. Enable WSL (If you're running a Windows)

Guide: <https://learn.microsoft.com/en-us/windows/wsl/install>

2. Install Git

Guide: <https://git-scm.com/install/linux>

3. Install VSCode (IDE)

Guide: <https://code.visualstudio.com/download>

4. Install gcloud CLI. (Google Cloud Toolkit)

Guide: <https://docs.cloud.google.com/sdk/docs/install#linux>

You may skip the installation instructions if the required developer tools are already present on your workstation.

Lab steps:

1. Clone the repo locally.

Once all Prerequisites are installed, clone the repo locally on your machine.

```
git clone https://<repo>/gdc-sandbox-workshop.git
```

2. Edit the .env file in vi on VSCode

```
cd ~/gdc-sandbox-workshop  
cp .env_sample .env
```

```
# edit .env for specifics  
vi .env
```

3. Run source .env

4. Establish the IAP tunnel

```
./sandbox.sh tunnel
```

5. RDP into your Sandbox jumpbox with your assigned sandbox user.

6. Clone the repo locally on sandbox.

Open a terminal and clone the repo locally.

```
git clone https://<repo>/gdc-sandbox-workshop.git
```

7. From your workstation copy the .env file to the sandbox.

```
./sandbox.sh env
```

8. Edit the project_config file with your landing zone info

```
vi projects_config.yaml
```

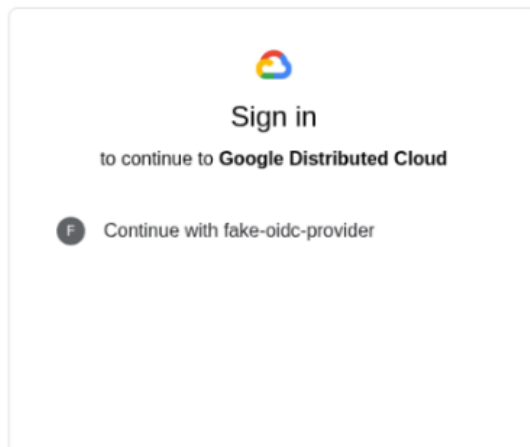
9. Copy the Landing Zone config file to the sandbox.

```
./sandbox.sh config
```

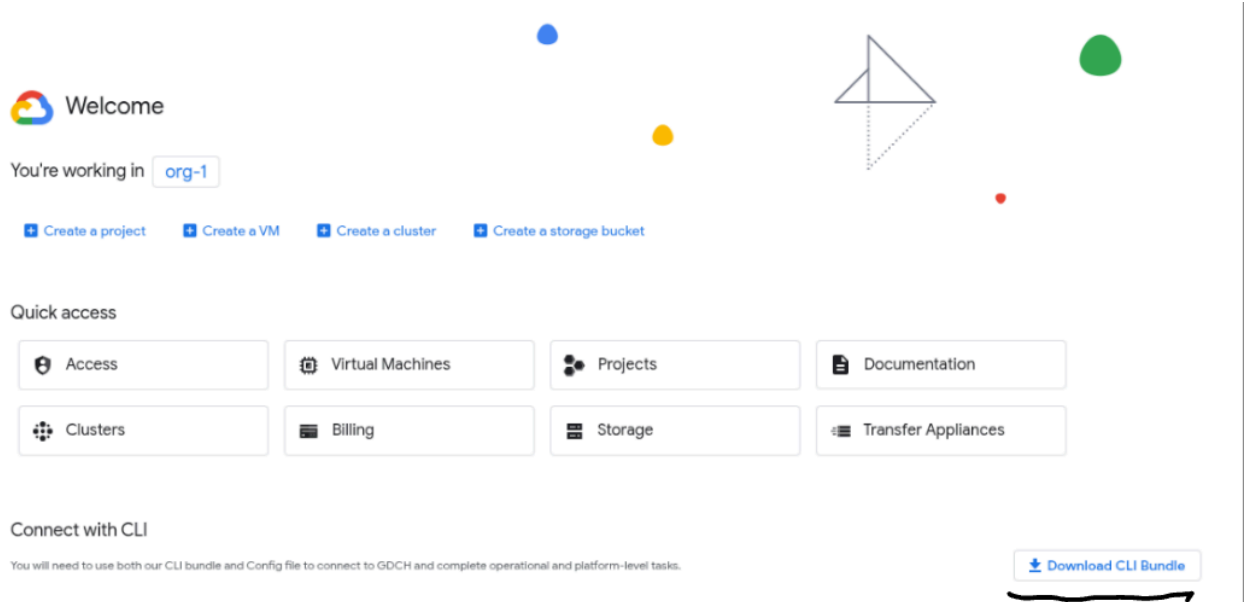
Landing Zone

1. Download the GDCloud toolkit from the GDC Console.

Navigate to the GDC Console: <https://console.org-1.zone1.google.gdch.test>



2. Login as Platform Admin
3. Download the CLI bundle from the landing page.



4. Run './000-install-gdcloud.sh' to install GDCloud toolkit
 5. source. env
 6. Run './login.sh' and login as "Platform Admin"
 7. Run './001-create-projects.py' to create your workloads Project
 8. Run './002-apply-role-bindings.py' Create your users and apply role bindings to your project
 9. Run './003-createharborproject.py' Create your project on the Harbor Instance
 10. Log into GDC with your sandbox username.
 12. Log into Harbor under the user-project, generate a user secret
- Log into the GDC Sandbox console with your login credentials and navigate to the Harbor menu.

Google Distributed Cloud

org-1 > user-project

Harbor Container Registry

Harbor Container Registry

Harbor is an open source registry that secures artifacts with policies and role-based access control, ensures images are scanned and free from vulnerabilities, and signs images as trusted.

Select a zone [Zone: zone1](#)

Harbor Instance

Status	Ready
Name	user-haas-instance

Harbor projects

A project in Harbor contains all repositories of an application. Images cannot be pushed to Harbor before a project is created. To view and manage your Harbor projects, [go to Harbor Instance](#)

Create A Harbor Project

Click on “Login via IAP provider”



Harbor

Search Harbor...

Harbor

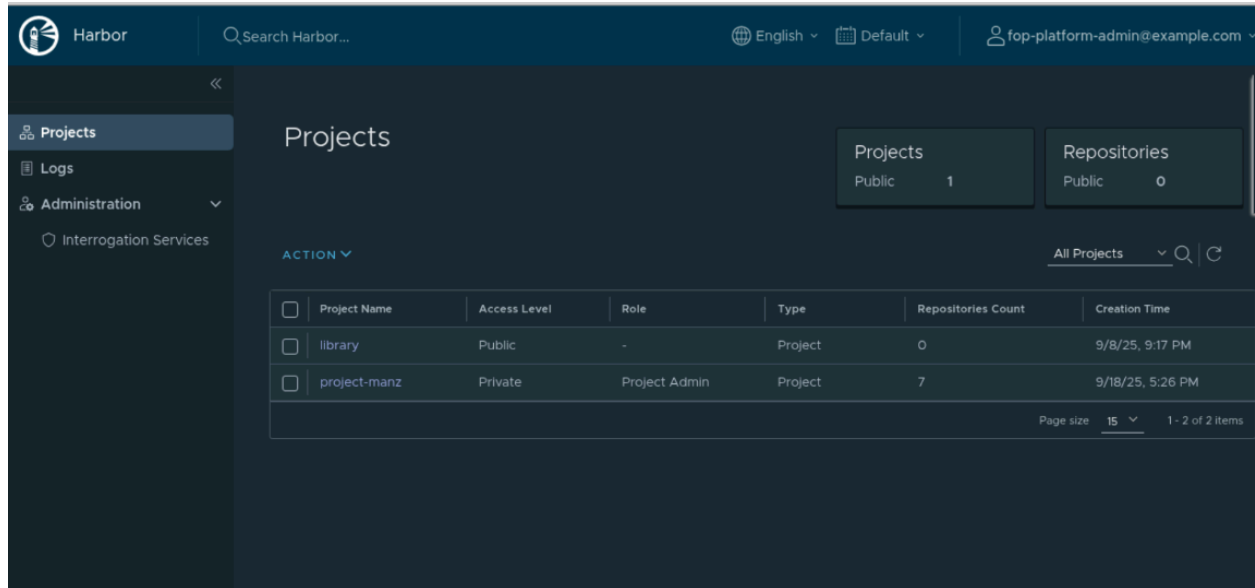
LOGIN VIA IAP PROVIDER

OR

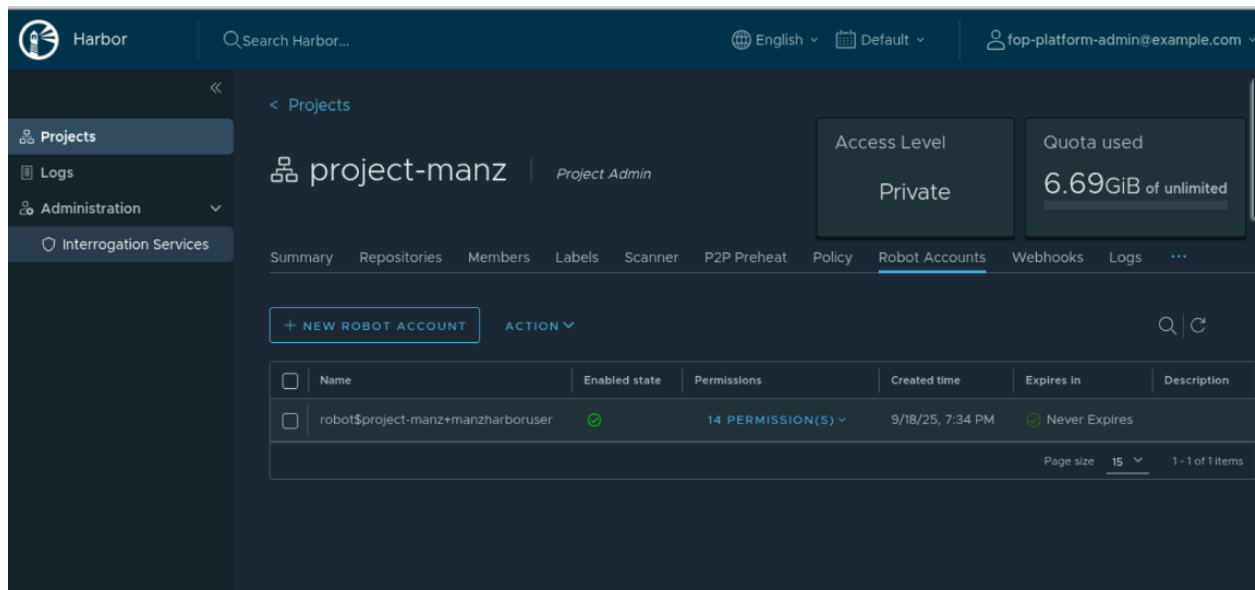
LOGIN VIA LOCAL DB

[More info...](#)

Click on your project



Click on “robot Accounts”



Create the Robot account and copy the secret.

project-manz

Project Admin

Access Level

Private

Summary

Accounts

Create Robot Account

Create a robot account for this project

Name ⓘ

test-user

Expiration time ⓘ

Never

-1

Description

Permissions

14 PERMISSION(S) ▾

CANCEL

ADD

✓

Created 'robot\$project-manz+test-user' successfully.

ⓘ

This is the only time to copy this secret. You won't have another opportunity

Name

robot\$project-manz+test-user

Secret

GNz9QDH6XvcC6avmxgrvKH3E42g6JWfV

EXPORT TO FILE

Update the harbor username and password in the .env file with the stored username and secret:

```
HARBOR_USERNAME='robot$project_name+harbor'
```

```
HARBOR_PASSWORD=nnnnnmmmmm111111
```

9. Run ./004-addharborsecret.py Create docker registry secret

Labs

Lab 1 - Deploy HTML Server

Follow the Lab Guide on Github: <https://<repo>/gdc-sandbox-workshop/blob/main/LAB-1.md>

Lab 2 - Deploy API Server

Follow the Lab Guide on Github: <https://<repo>/gdc-sandbox-workshop/blob/main/LAB-2.md>

Lab 3 - Deploy Elasticsearch

Follow the Lab Guide on Github: <https://<repo>/gdc-sandbox-workshop/blob/main/LAB-3.md>